REUBEN SAMSON RAJ

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EDUCATION

Ph.D. in Computer Science, University of Arkansas, Fayetteville, AR — Aug 2021 – Dec 2026 (Expected)
Dissertation: *Frameworks on Network Programmability and Abstraction for Securing and Strengthening Smart-Grids*
Advisor: Dr. Dong (Kevin) Jin

M.S. in Computer Engineering, University of Arkansas, Fayetteville, AR — Aug 2021 – Dec 2024

B.E. in ECE, College of Engineering, Guindy, Anna University, Chennai, India — Jun 2006 – May 2010

RESEARCH INTERESTS

Network Security, Programmable Networks, Smart-Grid/ICS Security, Cloud-Native Systems

WORK EXPERIENCE

Sr. Graduate Research Assistant — Aug 2021 – Present
University of Arkansas — Fayetteville, AR

Network Architecture Research Intern — Jun 2023 – Aug 2023
Nokia Bell Labs — Murray Hill, NJ

Software Engineer — Jun 2020 – Jul 2021
Cisco Systems — Bangalore, India

Member of Technical Staff — Oct 2018 – May 2020
Parallel Wireless — Bangalore, India

R&D Engineer — Apr 2015 – Oct 2018
Nokia Networks — Bangalore, India

Software Engineer — Apr 2012 – Apr 2015
Cisco Systems — Bangalore, India

Associate Software Engineer — Apr 2011 – Mar 2012
Accenture — Chennai, India

PUBLICATIONS

1. **Reuben Samson Raj**, Dong Jin. *A Framework to Evaluate PMU Networks for Resiliency Under Network Failure Conditions.* IEEE SmartGridComm 2022
2. Zhiyao He, Yanfeng Qu, Gong Chen, **Reuben Samson Raj**, Hui Lin, Dong Jin. *Towards Secure and Resilient Synchrophasor Networks Using P4 Programmable Switches.* IEEE GreenTech 2024
3. **Reuben Samson Raj**, Dong Jin. *Dynamic Data Driven Security Framework for Industrial Control Networks using Programmable Switches.* DDDAS 2024
4. **Reuben Samson Raj**, Dong Jin. *Leveraging Data Plane Programmability Towards a Policy-driven In-Network Security Framework for Industrial Control Systems.* ACM e-Energy 2025
5. **Reuben Samson Raj**, Dong Jin. *A Dynamic Data-Driven Framework for Enhancing Energy System Security and Resilience through Programmable Networks.* Book Chapter - DDDAS Handbook Vol. IV, Springer (Accepted-Under Production)

6. Sarvesh Bidkar, Rakesh Abbireddy, Maryam Amiri, **Reuben Samson Raj**, Anil Rana, Jeff McLaird, Paul Rea, Meryem Simsek, Jesse E. Simsarian. *Demonstration of Multi-Provider Network and Cloud Service Provisioning with Blockchain Smart Contracts.* ECOC 2025
7. Zhiyao He*, **Reuben Samson Raj***, Yanfeng Qu, Dong Jin. *Programmable Data Plane Approaches for ML-Based Anomalous Event Classification in Grid Sensor Networks.* IEEE ICC 2026 (* - Equal Contributions)
8. Yuexin Chen, **Reuben Samson Raj**, Xiaoyu Guo, Dong Jin. *Enhancing Security of Grid Control Systems through In-network Collaboration of Programmable Data Planes.* IEEE SmartGridComm 2026 (Accepted)
9. Haden Fowler, **Reuben Samson Raj**, Dong Jin. *Dynamic Data-Driven Cyber Defense for Industrial Control Systems using P4-Based Cross-Layer Telemetry.* DDDAS 2026 (Accepted)

AWARDS & HONORS

→ 2nd place - Best Paper Award, **DDDAS Conference**	2024
→ Student Travel Grant, **ACM e-Energy**	2025
→ Larry & Gwen Stephens Graduate Scholarship	2024
→ Reginald R. Barney & Jameson A. Baxter Graduate Scholarship	2023
→ Department of Electrical Engineering and Computer Science Scholarship	2022
→ You Amaze and You Inspire Awards, **Cisco**	2020
→ Best Feature Team Award, **Nokia**	2016
→ CAP Award, **Cisco**	2013

ACADEMIC SERVICE & VOLUNTEERING

→ Reviewer, *ACM Transactions on Modeling and Computer Simulation* (**TOMACS**), 2025, 2026

→ Reviewer, *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids* (**SmartGridComm**), 2026

→ Mentoring and guidance for undergraduate honor's thesis projects

→ Served as vendor liaison and key coding platform point-of-contact for regional high-school programming contests

SKILLS

Languages	Python, P4 (Tofino, bmv2), C, Bash
Networking	SDN, P4, 5G Core, EPC, TCP/IP, Mininet, Wireshark, iperf
Cloud & Orchestration	Kubernetes, Docker, OpenStack
Security Tools	ICS/OT (Modbus, DNP3), Network Security, IDA Pro, PE Studio
Tools & Frameworks	Linux, Git, FastAPI, OpenCV, NVIDIA GPU Stack

MEDIA REPORTS

news.uark.edu/articles/73095/second-best-paper-in-dynamic-data-driven-application-systems